

Space Exploration: Study Notes

The Solar System

Our solar system consists of the Sun, eight planets, five recognized dwarf planets, and countless smaller bodies such as asteroids, comets, and moons. The four inner planets — Mercury, Venus, Earth, and Mars — are rocky and relatively small. The four outer planets — Jupiter, Saturn, Uranus, and Neptune — are gas or ice giants with much larger diameters and are surrounded by ring systems and numerous moons.

Artificial Satellites

An artificial satellite is a human-made object placed into orbit around a celestial body, most commonly Earth. Satellites are used for communication, weather monitoring, navigation systems such as GPS, and scientific research. Low Earth orbit, roughly 160 to 2,000 kilometers above the surface, is favored for imaging satellites because of its proximity to the ground, while geostationary orbit at about 35,786 kilometers allows a satellite to remain fixed above a single point on Earth.

The Apollo Program

The Apollo program was a series of NASA missions in the 1960s and early 1970s with the goal of landing humans on the Moon and returning them safely to Earth. Apollo 11, launched in July 1969, achieved this goal when astronauts Neil Armstrong and Buzz Aldrin became the first people to walk on the lunar surface while Michael Collins orbited above in the command module. Six of the program's missions successfully landed crews on the Moon.

Rocket Propulsion

Rockets achieve thrust according to Newton's third law of motion: for every action there is an equal and opposite reaction. A rocket engine expels propellant mass at high velocity out of a nozzle, and the reaction force propels the vehicle in the opposite direction. The Tsiolkovsky rocket equation relates the change in velocity a rocket can achieve to its exhaust velocity and the ratio of its initial to final mass, and it explains why multi-stage rockets are more efficient than single-stage designs for reaching orbit.

Space Telescopes

Space telescopes observe the universe from above Earth's atmosphere, avoiding atmospheric distortion and absorption that limit ground-based observatories. The Hubble Space Telescope, launched in 1990, has captured detailed images of distant galaxies and played a key role in measuring the rate of expansion of the universe. The James Webb Space Telescope, launched in 2021, observes primarily in infrared light, allowing it to study the formation of the earliest galaxies and the atmospheres of exoplanets.

Mars Exploration

Mars has been a major focus of robotic exploration due to evidence that it once had liquid water on its surface. Rovers such as Curiosity and Perseverance analyze soil and rock samples to search for signs of past microbial life and study the planet's geology. Perseverance, which landed in 2021, also carries the Ingenuity helicopter, the first aircraft to achieve powered, controlled flight on another planet, and is caching rock samples for a future mission intended to return them to Earth.

The International Space Station

The International Space Station, or ISS, is a habitable artificial satellite in low Earth orbit that serves as a microgravity research laboratory. It is a collaborative project involving space agencies from the United States, Russia, Europe, Japan, and Canada, and has been continuously occupied by rotating crews of astronauts since November 2000. Experiments aboard the ISS study the effects of long-duration spaceflight on the human body, which is important for planning future missions to Mars.

Private Spaceflight

In recent decades, private companies have taken an increasingly large role in space exploration alongside government agencies. Reusable rocket technology, pioneered at scale by companies such as SpaceX, has significantly reduced the cost of reaching orbit by allowing rocket boosters to land and be flown again rather than being discarded after a single use. This shift has expanded commercial access to space for satellite launches, cargo resupply missions, and crewed flights to the ISS.